R20

Code No.: 20CIC04

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (CET) V Sem (Main & Backlog) Examination Dec 2023 – Jan 2024

Computer Networks

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		Draw a side view of a fiber optic cable and explain the components.	(3)	2	2
2		Explain single bit errors and burst errors with examples.	(3)	2	2
3		List and explain in brief about the functionalities of a Data Link Layer.	(3)	1	4
4		List the differences between TCP and UDP.	(3)	5	4
5		Explain Domain Name server (DNS).	(3)	6	2

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	Explain LAN, MAN and WAN technologies with suitable architecture diagrams.	(5)	2	2
	(b)	Explain in brief about the functionalities of session layer and presentation layer in OSI reference model.	(4)	1	2
		(OR)			
7	(a)	Explain circuit switching and packet switching	(5)	2	2
	(b)	Explain in brief about the functionalities of network layer and transport layer in OSI reference model.	(4)	1	2
8	(a)	Analyze CRC technique for error detection and correction with an example.	(7)	2	3
	(b)	A computer network uses polynomials over GF(2) for error checking with 8 bits as information bits and uses $\mathbf{x^3 + x + 1}$ as the generator polynomial to generate the check bits. Find how the message 01010010 is transmitted in this network.	(2)	2	4
		(OR)			
9	(a)	Discuss sliding window protocols.	(7)	2	4
	(b)	A network has a data transmission bandwidth of 20 X 10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. Find the minimum size of the frame (in bytes) in the network.	(2)	2	4
10	(a)	Explain link state routing with an example.	(5)	3	1
	(b)	Explain in brief about Distance Vector Routing with an example.	(4)	3	1
		(OR)			

11	(a)	Explain Dijkstra's shortest path routing algorithm with an example.	(5)	3	1
	(b)	What is flooding? What are the disadvantages of flooding? Explain measures to avoid/decrease the impact of flooding.	(4)	3	1
12	(a)	Sketch the TCP segment Header and explain all the components of the header.	(5)	5	3
	(b)	Explain Leaky bucket and Token bucket algorithms.	(4)	5	4
		(OR)			
13	(a)	Discuss TCP connection establishment and termination process with a neat diagram.	(5)	5	3
	(b)	Analyze the general principles of congestion control.	(4)	5	4
14	(a)	Explain the working of Electronic Mail.	(7)	6	2
	(b)	Explain the working of Hyper Text Transfer Protocol (HTTP).	(2)	6	2
		(OR)			
15	(a)	Explain the working of World-wide-web (WWW).	(7)	6	2
	(b)	What is a firewall and why is it used?	(2)	6	2

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